

Exercise 2.2: Advanced SQL - Date Functions

Note: Current date = 19-04-2026

---Q1: Add 6 months

```
SELECT emp_id,  
       name,  
       hire_date,  
       DATEADD(MONTH, 6, hire_date) AS hire_plus_6_months  
FROM Employees;
```

emp_id	name	hire_date	hire_plus_6_months
1	Alice	2020-01-15	2020-07-15
2	Bob	2021-06-10	2021-12-10
3	Charlie	2023-03-22	2023-09-22

---Q2: Age in days

```
SELECT student_id,  
       name,  
       DATEDIFF(day, dob, CURRENT_DATE) AS age_in_days  
FROM Students;
```

student_id	name	age_in_days
101	Maya	7553
102	Ethan	7810
103	Sierra	7339

---Q3: Days remaining to events

```
SELECT event_id,  
       event_name,  
       DATEDIFF(day, CURRENT_DATE, event_date) AS days_remaining  
FROM Events;
```

event_id	event_name	days_remaining
1	Seminar	-673
2	Workshop	-230
3	Hackathon	-454

Q4: days between issue & due date

```

SELECT invoice_id,
       issue_date,
       due_date,
       DATEDIFF(DAY, issue_date, due_date) AS days-between
FROM Invoices;

```

invoice_id	issue_date	due_date	days-between
S01	2025-03-10	2025-03-25	15
S02	2025-04-01	2025-04-15	14
S03	2025-04-10	2025-04-20	10

---Q5 Format start date

```

SELECT course_id,
       name,
       TO_CHAR(start_date, 'Month YYYY') AS formatted_date
FROM Courses;

```

course_id	name	formatted_date
201	SQL Basics	May 2025
202	Python	June 2025

---Q6 DATE_FROM_PARTS() builds full date from separate columns

```

SELECT member_id,
       DATE_FROM_PARTS( start_year, start_month, start_day) AS full_start_date
FROM Membership;

```

member_id	full_start_date
1	2023-05-10
2	2022-11-25

---Q7 : Extend by 1 year

```

SELECT sub_id,
       plan,
       DATEADD(YEAR, 1, renewal_date) AS extended_renewal_date
FROM Subscriptions;

```

sub_id	plan	extended_renewal_date
11	Basic	2026-01-01
12	Premium	2025-03-15

---Q8 Days since order

```

SELECT order_id,
       order_date,
       CURRENT_DATE AS today_date,
       DATEDIFF(DAY, order_date, CURRENT_DATE) AS days_since_order
FROM Orders;

```

order_id	order_date	today_date	days_since_order
1001	2025-04-15	2026-04-19	369
1002	2025-04-10	2026-04-19	374

---Q9 DATE PART() / EXTRACT() pulls specific part from date

```
SELECT training_id,  
       topic,  
       EXTRACT (YEAR FROM training_date) AS training-year  
FROM Trainings;
```

training_id	topic	training-year
1	Safety	2025
2	Compliance	2025

--Q10 Extract hour & minute

```
SELECT post_id,  
       title,  
       EXTRACT (HOUR FROM published_on) AS hour-published,  
       EXTRACT (MINUTE FROM published_on) AS minute-published  
FROM Blog_Posts;
```

post-id	title	hour-published	minute-published
1	SQL Tips	10	15
2	Data Cleaning	16	45

--Q11 Days until expiry

```
SELECT driver_id,  
       license_expiry,  
       DATEDIFF (DAY, license_expiry, CURRENT_DATE) AS days-left  
FROM Drivers;
```

driver-id	license_expiry	days-left
801	2025-08-10	-252
302	2023-12-31	-840

--- Q12 seconds since message sent

```
SELECT message_id,  
       sent_timestamp  
       current_timestamp  
       DATEDIFF(SECONDS, sent_timestamp, CURRENT_TIMESTAMP) AS seconds_since_  
       sent  
FROM Messages;
```

message-id	sent_timestamp	current_timestamp	seconds_since_sent
1	2025-04-19 09:32:45	2026-04-19 14:30:45	31554000
2	2025-04-18 23:59:59	2026-04-19 14:30:45	31559446

--- Q13 Add 15 days

```
SELECT return_id,  
       return_date  
       DATEADD(DAY, 15, return_date) AS restock_date  
FROM Returns;
```

return-id	return_date	restock_date
901	2025-04-05	2025-04-20
902	2025-04-01	2025-04-16

--- Q14 convert date ^{using} (TO-DATE) [converts string to date]

```
SELECT assign_id,  
       TO-DATE(assigned_on, 'YYYY-MM-DD') AS assigned_on_date  
FROM Assignments;
```

assign-id	assigned_on_date
1	2025-03-01
2	2025-03-05

--- Q15 : Formatting time (convert ^{date} ~~string~~ to ^{string} ~~date~~)

SELECT meeting_id,

TO_CHAR(scheduled_time, 'Month DD', 'YYYY' "at" HH:MI AM)

AS formatted_meeting_time

FROM Meetings

meeting_id	formatted_meeting_time
1	April 19 at 02:00 PM
2	Apr 19 at 09:30 AM